PhyriadFG

External-capture, multi-GPU frame generation — architecture

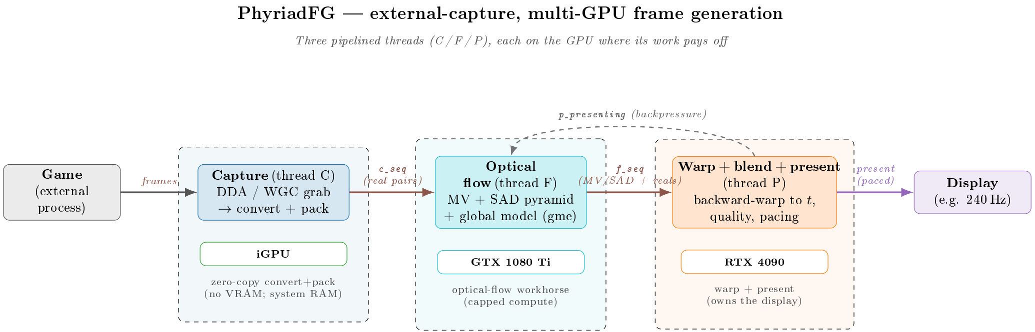
Student-built, LLM-assisted · `0.1.0-experimental`

What PhyriadFG is

- A **frame generator** that runs **outside** the game: it captures the already-rendered output, estimates motion between real frames, synthesizes intermediate frames, and presents them.
- **No engine access** — no motion vectors, no depth, no hooks. Only the final image.
- **Multi-GPU by design**: capture+convert on the iGPU, optical flow on a second GPU, warp and present on the primary GPU — each role where its work pays off.
- One executable, one build (`build.bat` → `phyriad_fg.exe`), LTO on.

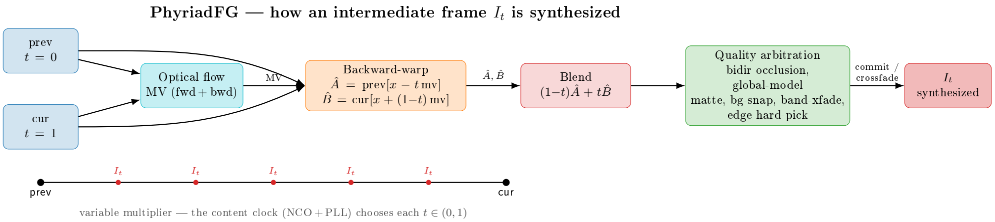
Honest scope: an external capture+interpolation tool. It cannot exceed the display refresh without injection or a driver, and it shares one or more GPUs with the game.

Architecture — three threads, three GPUs



Threads communicate through `FgContext` (SPSC rings + the sequence counters `c_seq` / `f_seq` / `p_presenting`); the present thread back-pressures the flow thread.

How a frame is generated



Backward-warp both reals to phase t, blend, then let the quality layers arbitrate the hard cases (occlusion, disocclusion). The content clock chooses t — the multiplier is variable.

Single source of truth — the config resolver

Flags parse into a `Config`; then `resolve_config()` does, in *one* place:

1. **Dependency cascades** (order-independent): requires / disables / auto-enables.
2. **Derived predicates** (`Config::Derived`): the single owner of cross-layer decisions — e.g. `field_to_warp` = "the iGPU contour field must reach the warp". The create/import/upload sites read *this*, instead of each re-deriving its own gate (which is how they drift).
3. **A startup self-audit**: warns when a requested flag can have no effect (no silent no-ops).

Idempotent + re-callable: a runtime degrade re-resolves in the same place. **To add a capability that reads a shared resource, register it in the predicate — the consumer sites never change.**

Per-tick control — the governor control word

Under GPU saturation an optional governor graduates down non-essential work so the FG keeps generating. The decision is **one control word, one owner**:

- `governor_floor_for_util()` — the single decode of utilization $\rightarrow$ tier floor.
- `g_gov_floor` (atomic) — **computed and published by the present thread** (it owns the utilization reading + hysteresis), **read by the flow thread** as `max(cpu_ladder, floor)`.
- One producer, one consumer, advisory + lock-free. Governor off or GPU cool $\Rightarrow$ floor 0 $\Rightarrow$ no-op.

One decode site, never two threads interpreting utilization independently.

Quality layers (opt-in, composable)

On top of the warp+blend core, each addressing a specific failure mode of external-capture interpolation:

- bidirectional occlusion classification
- global affine motion model (gme) + matte
- background snap (kills disocclusion "gravity")
- band crossfade
- edge-gated hard-pick
- sub-pixel & candidate-selected motion vectors
- object / scene holons (motion-inheritance repair)
- content-clock pacing, ASW extrapolation

Each is a flag with a documented default; off paths are byte-identical. Composition is resolved centrally (previous slide).

Build & repository layout

```
build.bat -> build\phyriad_fg.exe        (MSVC + Vulkan SDK + Ninja, LTO)

src/core/      orchestrator: main, fg_context, globals, device, vk_util
src/cli/       Config + parsing + resolve_config (the resolver)
src/capture/   run_capture  [thread C]  DDA / WGC, convert/pack
src/flow/      run_flow     [thread F]  optical flow, gme, holons
src/warp_blend the warp-at-presenter + field/fill
src/present/   run_present  [thread P]  warp, quality, pacing, present
framework/     pillars: render/present, render/vulkan, hal, topology, schema
shaders/       the .comp shaders        ui/  Tauri launcher
```

Honest scope

- **Is**: an external capture + interpolation + present pipeline, multi-GPU, with a quality stack for the hard cases and a launcher UI.
- **Is not**: an in-engine FG (no motion vectors / depth); it does not beat the display refresh without injection or a driver; it shares the GPU(s) with the game.
- `0.1.0-experimental`. Student-built, LLM-assisted. Numbers are stated only where measured.